MagiAttention

Distributed attention for ultra-long, heterogeneous-mask training

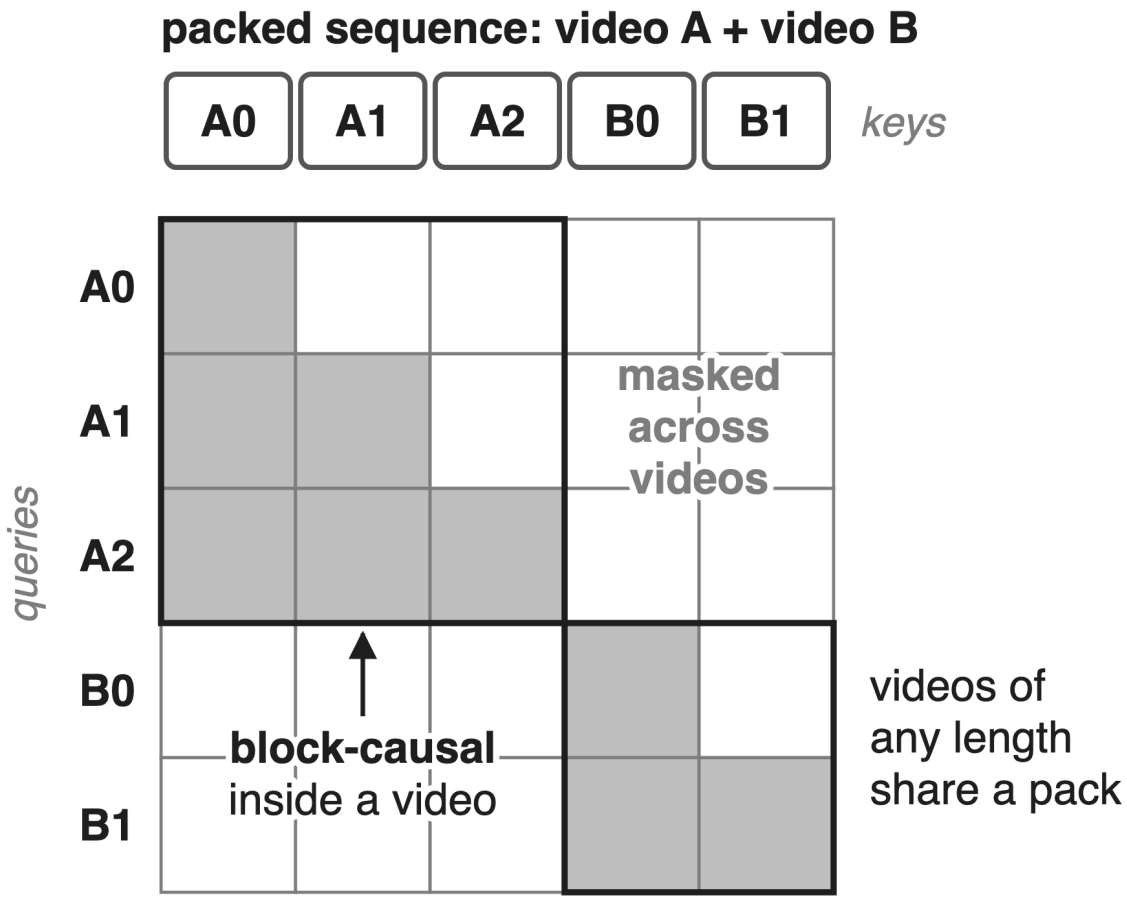
Sand AI · MAGI-1

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Frameworks & Platform Stacks · github.com/SandAI-org/MagiAttention

MAGI-1 trains video at up to 4M tokens

- **24B** autoregressive video model, contexts up to **4M** tokens
- 24-frame chunks attend to all earlier chunks
- Online packing mixes videos of any length



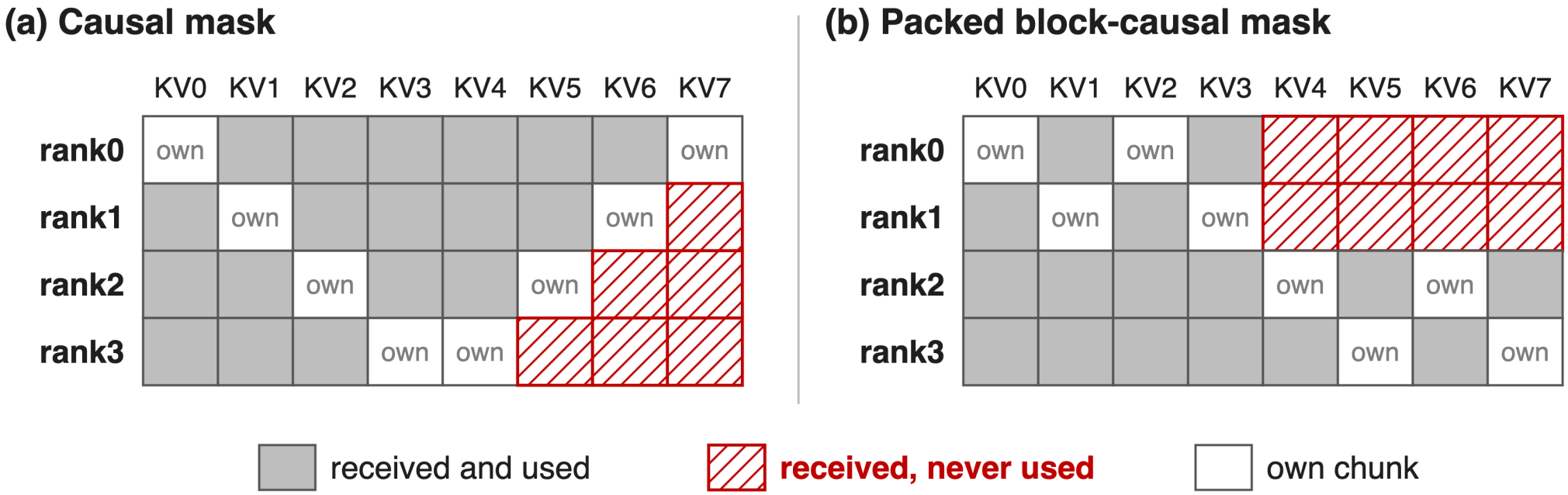
Packed masks cut ring attention ≈18×

≈18×

lower per-GPU throughput for ring P2P attention on a packed causal mask (≈29 TFLOPs/s) than on a full mask (≈513), 64 H100s, forward

MagiAttention CP benchmark blog, distributed full and varlen causal charts (values read from the charts; ratio derived)

Ring CP ships KV by position, not need

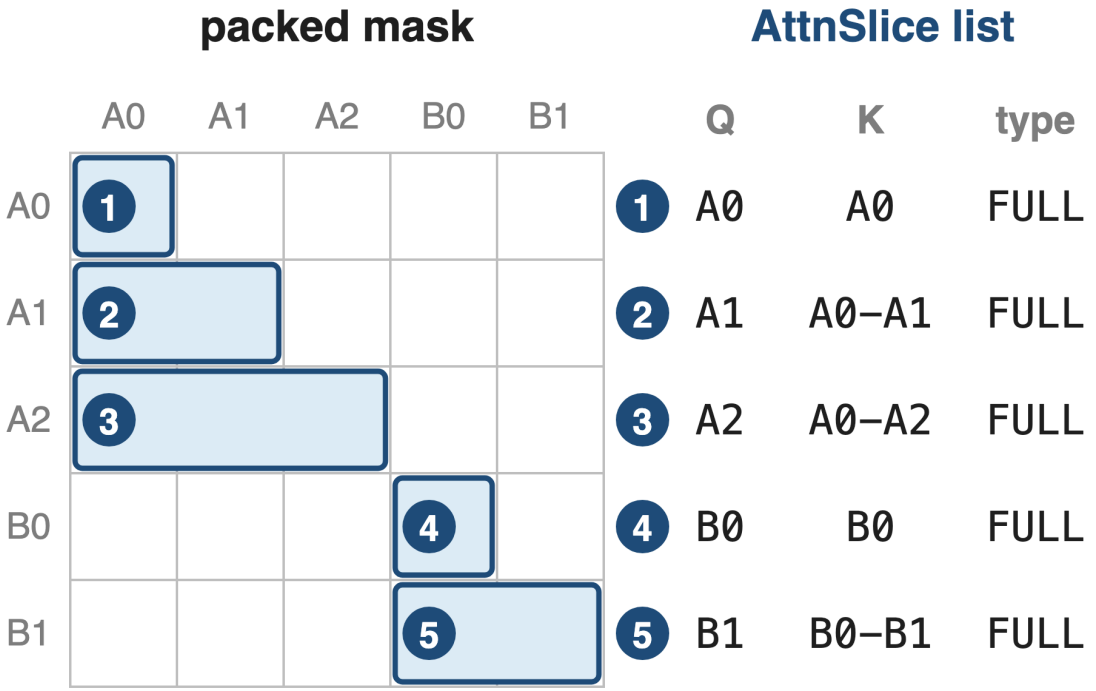


4 ranks, 8 chunks, 2 per rank: ring P2P delivers every remote chunk, and 25% (a) or 33% (b) of it is never read

Zigzag balancing fits the causal mask only; Ulysses needs heads divisible by CP size

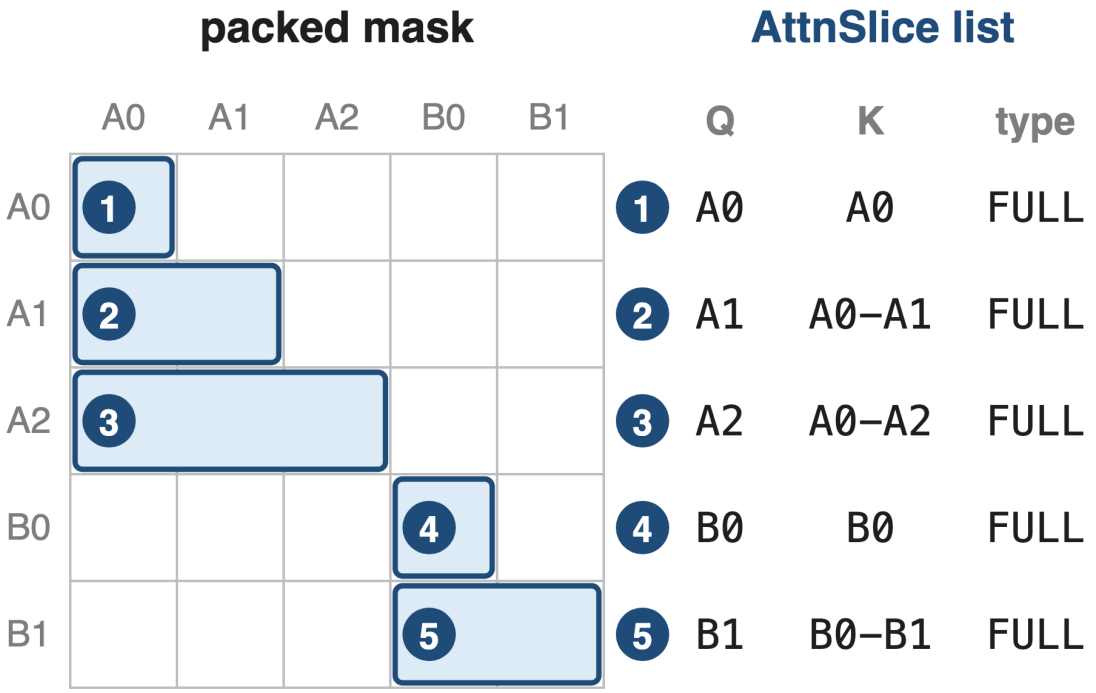
Insight: plan CP from the mask itself

- AttnSlice: Q range, K range, mask type
- Still a slice list after any re-sharding



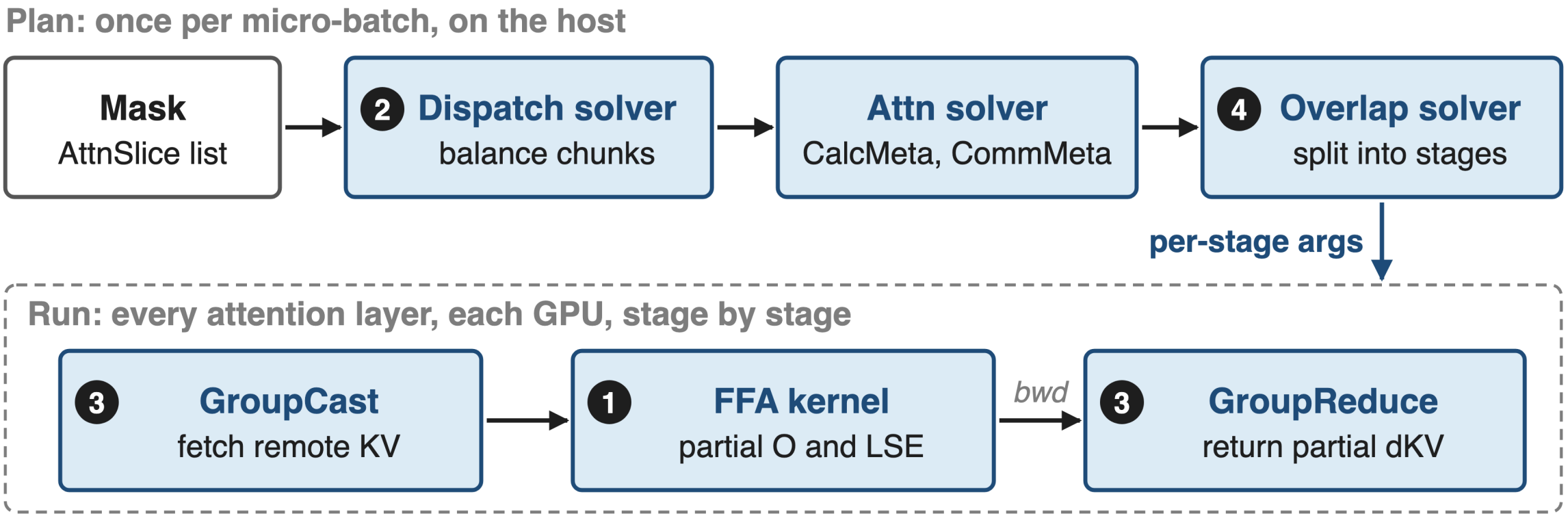
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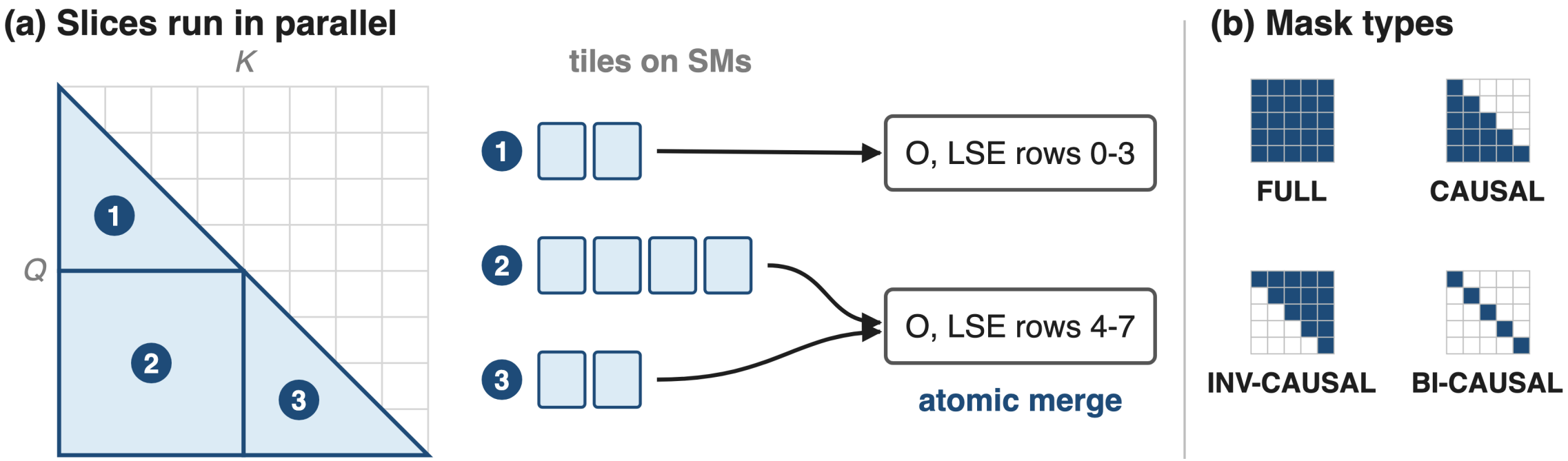
The slice list tells each rank what to compute and which remote KV it needs, before the step runs

MagiAttention compiles the mask into a plan



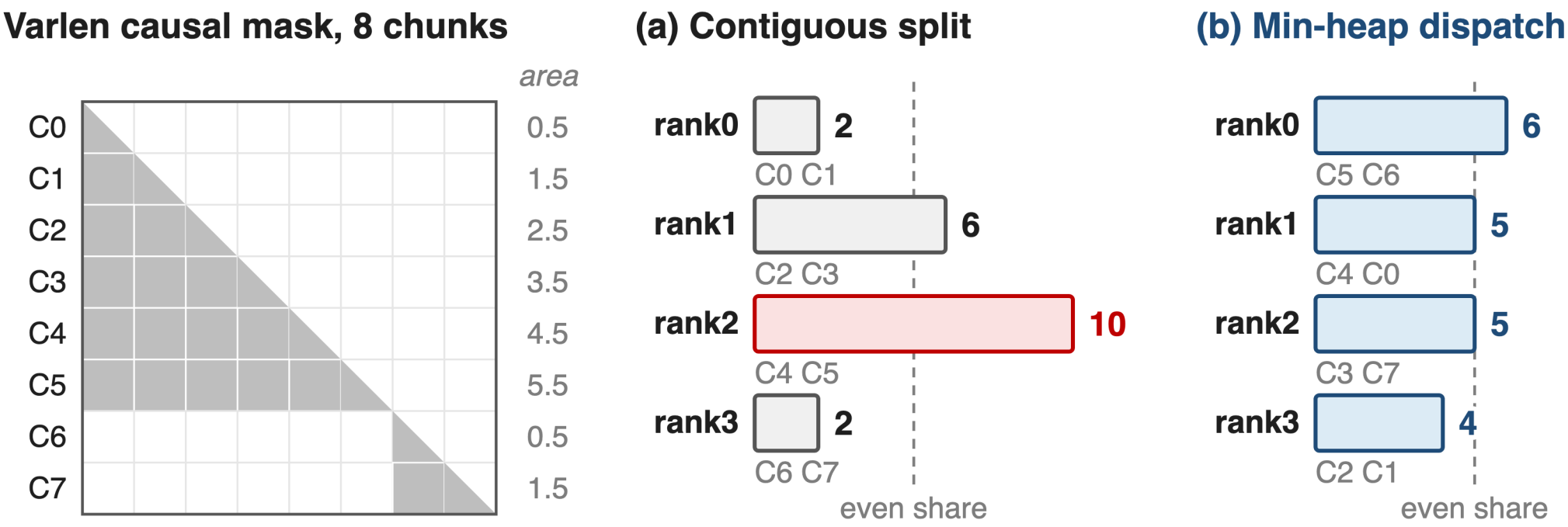
API: magi_attn_flex_key, dispatch, calc_attn, undispatch; examples for Megatron-LM, PyTorch FSDP and HF Transformers

❶ FFA computes any slice list in one kernel



Built on FA3 for Hopper; on Blackwell, FFA_FA4 turns slices into per-row key intervals on a forked FA4

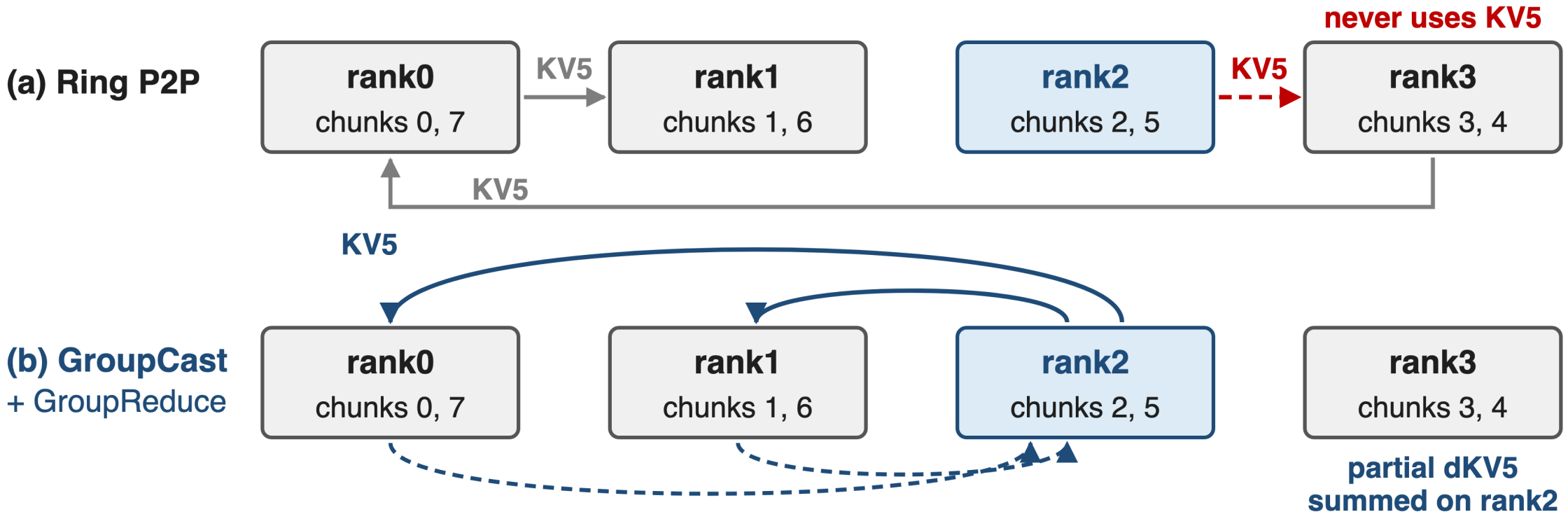
❷ Dispatch balances mask area, not tokens



Analytic example. Min-heap greedy: largest chunk first, to the least-loaded rank with room; 2 chunks per rank

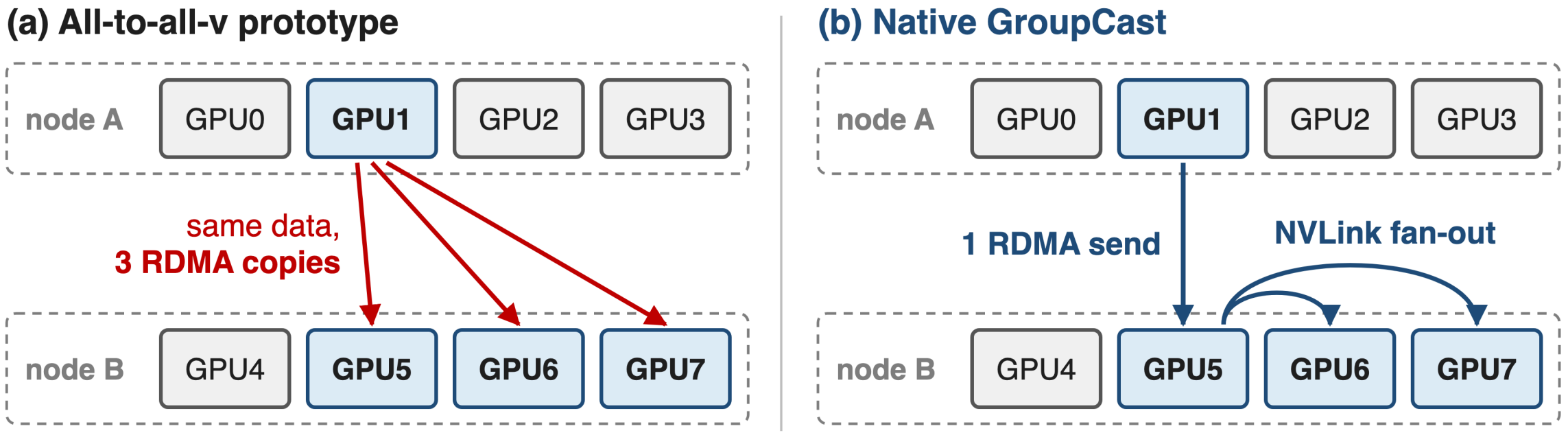
The slowest rank drops from 10 to 6 work units; token-wise layers stay balanced

❸ GroupCast sends KV only where needed



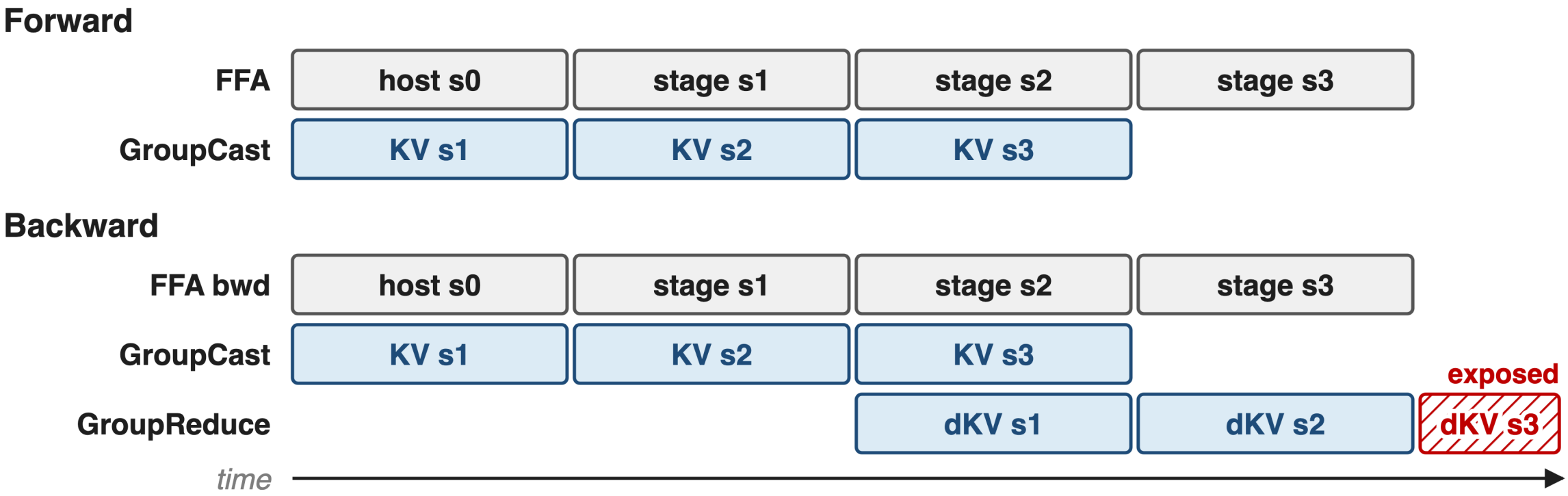
First built on all-to-all-v: zero redundancy, but extra copy kernels and one buffer per peer

❸ Native kernels turn RDMA copies into NVLink



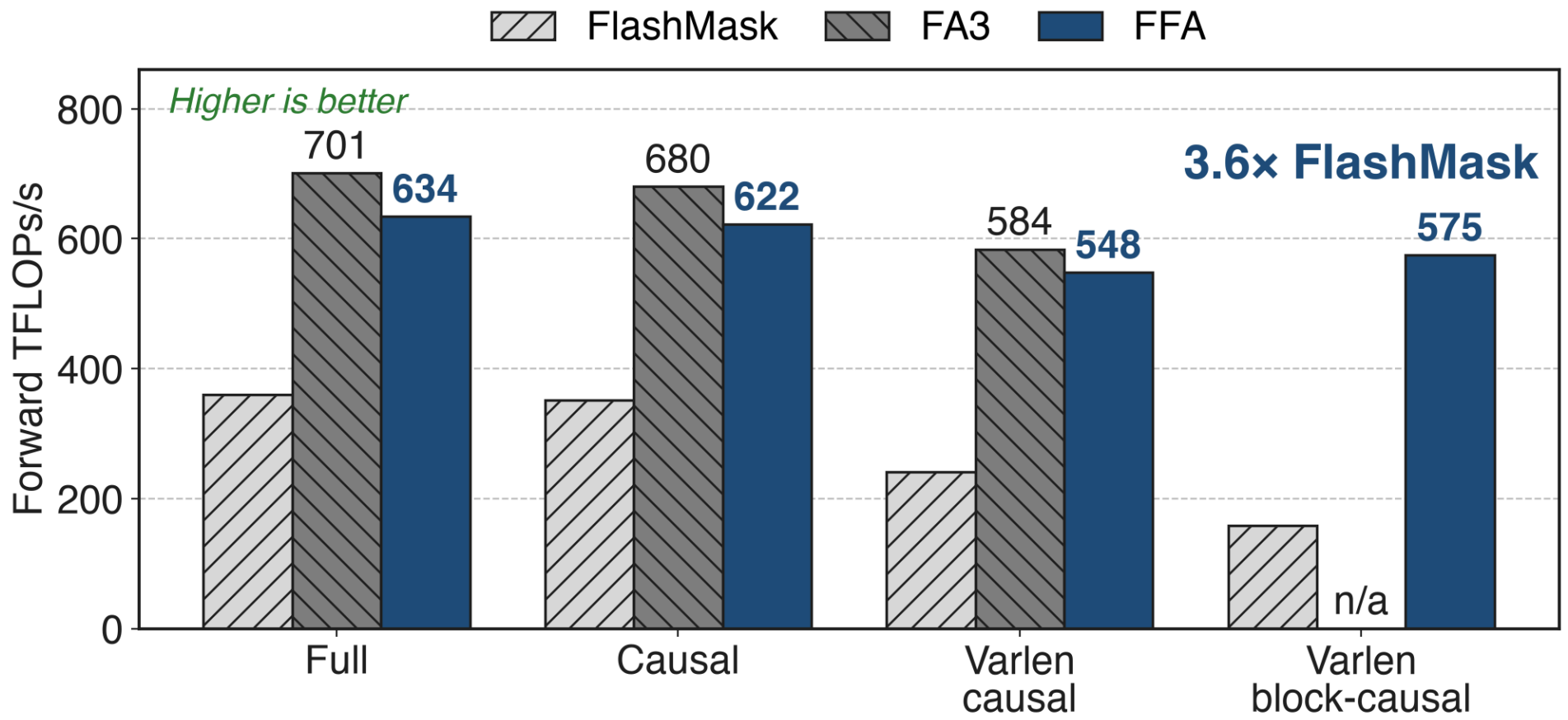
Built on DeepEP's dispatch and combine kernels (deck 24): no extra copy kernels, LSE merge inside the kernel

❹ Stages hide KV traffic under FFA



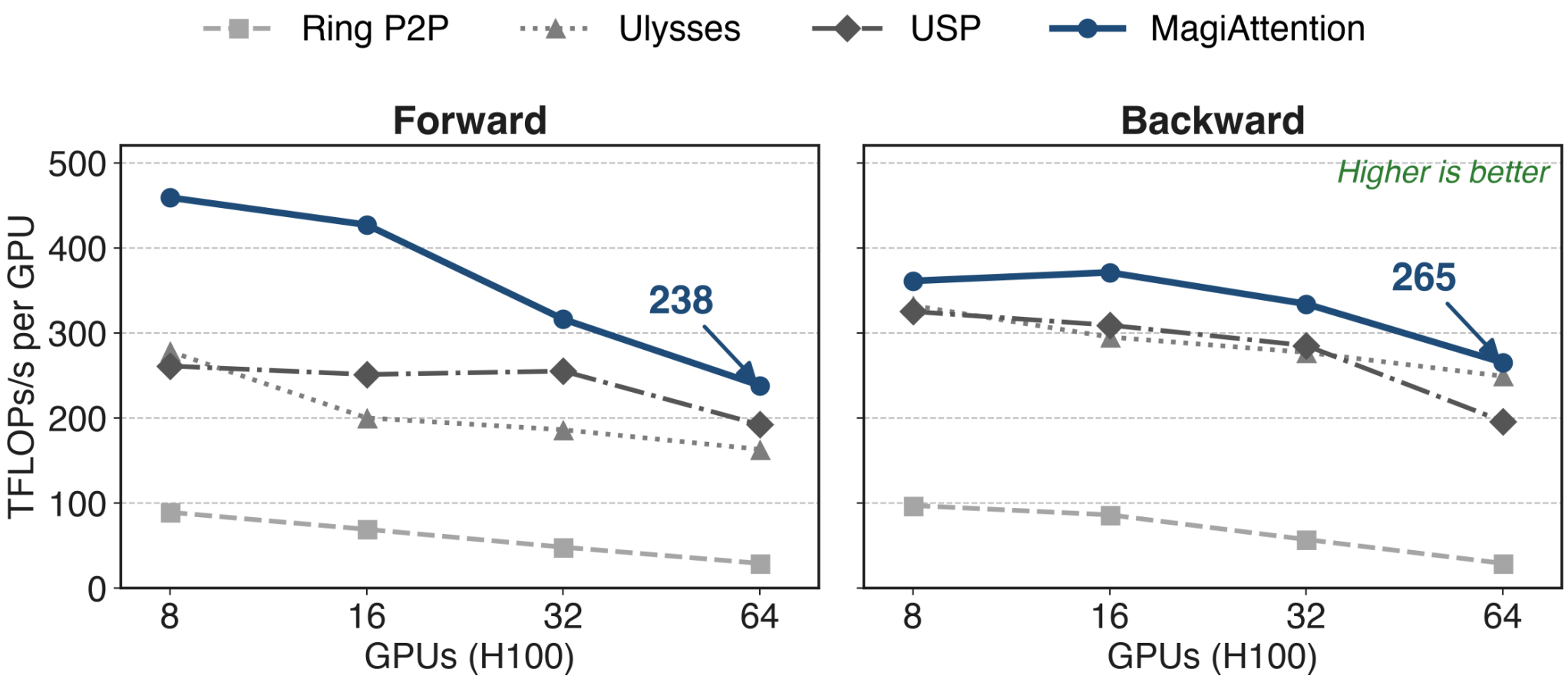
overlap_degree (1-4) is tuned by hand; a KernelBarrier lets comm kernels take SMs before FFA

FFA: ≈90% of FA3, 3.6× FlashMask



H100, BF16, GQA 64:8:8, head dim 128, 32K tokens, forward; values read from the MagiAttention benchmark charts

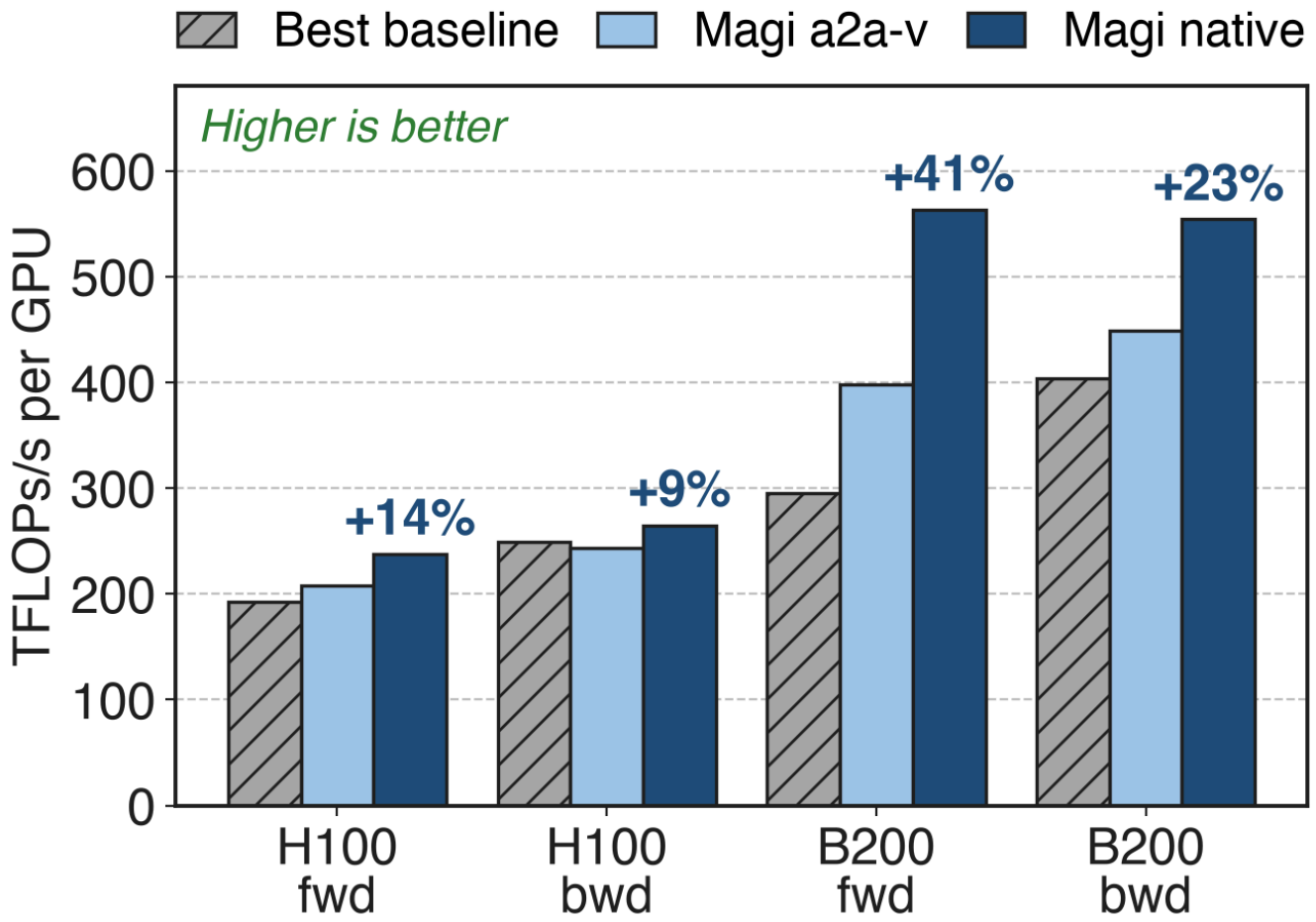
64 H100s: 8× ring P2P, 1.06-1.24× best baseline



Varlen causal mask, 8K tokens per GPU (64K-512K total); all baselines on FA3; values read from the benchmark charts

Native kernels add up to 41% at 64 GPUs

- Over all-to-all-v: **+9-14%** H100, **+23-41%** B200
- B200: **1.4-1.9×** the best baseline
- B200 baselines on cuDNN, Magi on FFA_FA4



The cost: hand tuning, IBGDA, no full-mask edge

Limits

- Per-GPU throughput still falls: **459** → **238** (8 → 64 H100)
- Full mask: backward ≈5% behind USP
- Vendor benchmarks, experimental flags on

Prerequisites

- IBGDA for internode native kernels
- overlap_degree tuned by hand
- Mask fixed per micro-batch; tokens permuted

Summary

- Ring CP on packed masks: **≈18×** below full-mask throughput
- **Insight**: plan compute and traffic from the mask's AttnSlices
- FFA at **≈90%** of FA3 · area-balanced dispatch · GroupCast · staged overlap
- 64 H100s: **8×** ring P2P; 64 B200s: **1.9×** best baseline

https://github.com/SandAI-org/MagiAttention